Paper-Ready Setup, Ablation, and Trigger-Generalization Results

Week 11: the experimental parameter card, the defense-layer ablation, and the trigger-generalization comparison

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Everything in this report is produced by one notebook (11_paper_tables.ipynb, submitted separately) and exported to results/ as CSVs and a PNG; nothing is transcribed by hand. The pipeline, model, attack, and the adopted D2 defense configuration are carried over unchanged from the Week 10 validation notebook. All experiment tables report mean +/- standard deviation over three federated seeds (42, 7, 123) on a fixed data split, with backdoor lift always paired within-seed against that seed's own honest baseline.

1. Attack setup / experimental parameter table

Group	Parameter	Value
Data	Dataset	Aissou et al. 2022, GPS spoofing on UAS (simplified 2D feature map)
	Subsample used	150,000 rows (90,000 benign, 60,000 spoofed; ratio 3:2)
	De-duplication	on the 10 model features, after dropping PRN/RX/TOW (Week 10 leak fix)
	Client training pool	114,000 rows, scaler fit on this pool only
	Server root / challenge set	6,000 rows, disjoint from train and test (hash-verified)
	Test set	30,000 rows (20%), stratified, fixed across seeds
	Input features	10: DO, PD, CP, EC, LC, PC, PIP, PQP, TCD, CN0
Federation	Clients	10 (IID split, equal benign/spoofed per client)
	Malicious clients	2 (IDs C9, C10)
	FL rounds	12
	Local epochs per round	3
	Batch size / optimizer	512 / Adam, lr 1e-3
	Local validation fraction	0.15
	Random seeds	data split fixed at 42; federated seeds [42, 7, 123] (partition, poison draw, init, batching)
Model	Architecture	10-64-32-16-1 MLP, ReLU, dropout 0.2, BCE-with-logits
	Parameters	3,329 (13.0 KB in float32)
Attack	Poison ratio	40% of each attacker's spoofed rows, relabeled authentic
	Trigger feature (default)	CN0
	Trigger value	benign 75th percentile: 46.706 raw (+0.738 standardized)
	Update scaling factor	3.0 (model replacement)
	Fake reported accuracy	0.99 (inflation variant only)
Defense	Aggregation methods compared	FedAvg; accuracy-weighted FedAvg; coordinate-wise median; trust-weighted FedAvg; trust-scaled median (full)
	Probe features	8 features with Cohen's d >= 0.05: DO, TCD, CN0, PC, LC, EC, CP, PD
	Gate sharpness beta	1.0
	Suspicion dead-zone tau	2.0 (MAD units)
	Trust smoothing EMA	0.5
	Flag threshold (reporting)	trust < 0.05 (half of uniform 1/N = 0.1)

Table 1. The full reproducibility card, assembled from the notebook's live variables rather than typed in. The one data-dependent value, the trigger value, is pinned in both raw and standardized units.

Insight: a reader can reproduce the full experiment from this table alone: the 150,000-row subsample and its 3:2 benign/spoofed ratio, the 6,000-row server root set (hash-verified disjoint from train and test), the 10-client IID federation with C9 and C10 compromised, the 3,329-parameter detector, the attack knobs, and the D2 defense constants.

2. Three-seed ablation: does each defense layer matter?

Six configurations plus one robustness check, all against the same attack (40% poison, CN0 trigger, update scaling 3.0), all across three seeds. Median only and trust only isolate the two layers of the full defense.

Method	Clean Accuracy	Spoofing Recall	BSR	Backdoor Lift
Honest FedAvg (no attack)	0.7109 +/- 0.0021	0.5287 +/- 0.0124	0.6368 +/- 0.0138	+0.0000 +/- 0.0000
Attack (FedAvg)	0.6928 +/- 0.0032	0.3618 +/- 0.0091	0.8825 +/- 0.0157	+0.2457 +/- 0.0023
Attack + inflation (Acc-Weighted)	0.6900 +/- 0.0047	0.3535 +/- 0.0158	0.9404 +/- 0.0101	+0.3036 +/- 0.0131
Median only	0.7098 +/- 0.0038	0.4993 +/- 0.0143	0.7009 +/- 0.0084	+0.0641 +/- 0.0064
Trust only	0.7114 +/- 0.0017	0.5311 +/- 0.0137	0.6404 +/- 0.0103	+0.0037 +/- 0.0042
Full defense (trust + median, D2)	**0.7142 +/- 0.0027**	**0.5546 +/- 0.0177**	**0.6114 +/- 0.0315**	**-0.0253 +/- 0.0178**
Full defense vs attack + inflation	0.7142 +/- 0.0027	0.5546 +/- 0.0177	0.6114 +/- 0.0315	-0.0253 +/- 0.0178

Table 2. Ablation, mean +/- std over seeds 42, 7, 123 (n = 3). BSR is the backdoor success rate on the triggered test set; backdoor lift is BSR minus the same seed's honest baseline. The highlighted row is the adopted configuration.

Insight

- The attack works and inflation makes it worse: lift +0.2457 under FedAvg, +0.3036 when fake accuracy reports buy the attackers extra weight.
- Median only helps but leaves a real backdoor effect: +0.0641 +/- 0.0064, about a quarter of the attack survives. Two attackers in ten can still drag a coordinate-wise median.
- Trust only removes nearly all of the lift (+0.0037 +/- 0.0042) because the probes identify the attackers directly. On the lift metric alone it is statistically close to the full defense, and we say so.
- The full defense is the only configuration that drives lift negative (-0.0253 +/- 0.0178) and it posts the best clean accuracy (0.7142) and spoofing recall (0.5546) at the same time. The median layer is the safety net if the trust scores are ever wrong in a round, a failure mode trust-only cannot bound.

Robustness check (last row): against attack + inflation the full defense produces numerically identical results, and the reason is structural rather than lucky. Trust comes entirely from server-side probing; the self-reported accuracy that inflation manipulates is never consulted by the full defense.

3. Trigger generalization, including a mixed trigger

Four trigger settings, each run undefended and under the full defense at the identical D2 configuration; nothing is retuned per trigger. PD is the weakest probed feature and the genuine stress test; the mixed CN0+TCD trigger is a pattern the defense was never designed around.

Trigger	Cohen's d	Attack BSR	Attack lift	Defended BSR	Defended lift	Honest FP rate
CN0	0.288	0.8825 +/- 0.0157	+0.2457 +/- 0.0023	0.6114 +/- 0.0315	-0.0253 +/- 0.0178	0.3%
TCD	0.302	0.8428 +/- 0.0075	+0.2384 +/- 0.0729	0.5419 +/- 0.0261	-0.0626 +/- 0.0512	0.7%
PD	0.152	0.9168 +/- 0.0069	+0.0909 +/- 0.0106	0.8071 +/- 0.0175	-0.0188 +/- 0.0150	0.3%
CN0+TCD	0.288 + 0.302	0.9816 +/- 0.0051	+0.1933 +/- 0.0664	0.7600 +/- 0.0683	-0.0282 +/- 0.0173	0.7%

Table 3. Trigger generalization at fixed D2, mean +/- std over seeds 42, 7, 123 (n = 3). BSR and lift are measured on each setting's own triggered test set against the same per-seed honest model. Honest FP rate is the fraction of honest client-rounds with trust below half of uniform.

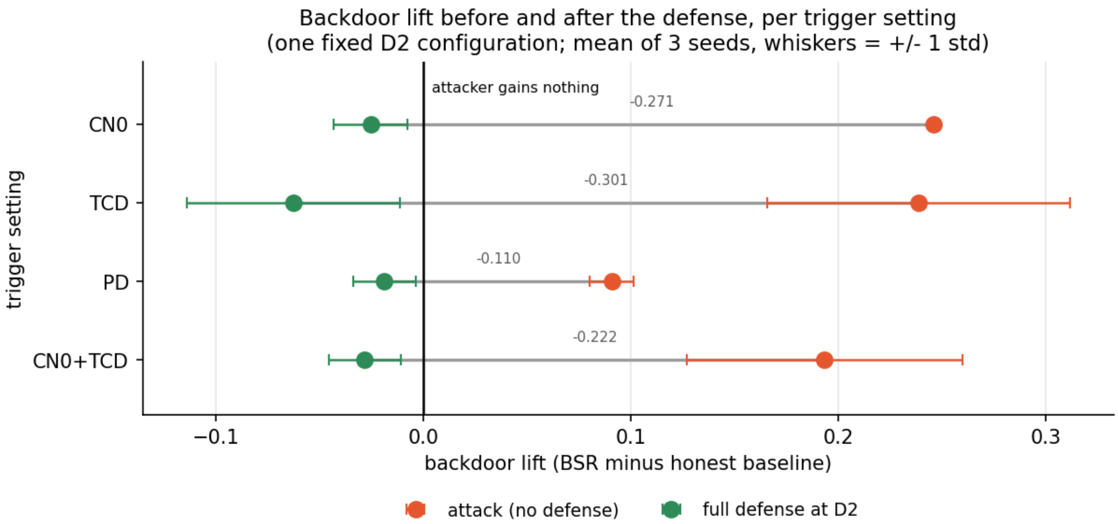

Figure 1. Backdoor lift before (orange) and after (green) the defense, per trigger setting. Each connector spans one trigger's undefended and defended lift, so its length is the backdoor the defense removes; markers are means over three seeds and whiskers are +/- 1 std. The black line at zero marks "the attacker gains nothing."

Insight

- The attack carries a viable backdoor in every setting (+0.09 to +0.25 lift), so each row is a real test.
- The defense drives lift below zero in every setting: CN0 -0.0253, TCD -0.0626, PD -0.0188, mixed -0.0282. Honest false positives stay between 0.3% and 0.7% of client-rounds throughout.
- Honest observations: the seed spread on the TCD and mixed attack rows is wide (std 0.05 to 0.07), and the mixed trigger has the highest raw BSR (0.9816) but not the highest lift, because stamping two features raises the honest baseline as well.
- Together with the Week 10 five-feature sweep, this supports the claim that the defense does not require knowing the exact trigger feature, within the discriminative (probed) feature set.

4. Adaptive (defense-aware) attacker stress test

The ablation and generalization attackers are fixed: they poison and scale without reacting to the defense. Here the attacker knows exactly how the coordinator probes and trains to evade it, adding an evasion term that keeps its model predicting spoofed on probe-style slices so it looks like the honest cohort. The evasion strength lambda is swept from 0 (the ordinary attacker) upward.

Evasion strength (lambda)	Undefended lift	Defended lift (D2)	Mean attacker trust
0.0	+0.2457 +/- 0.0023	-0.0253 +/- 0.0178	0.0001 +/- 0.0001
2.0	-0.1433 +/- 0.0439	-0.0534 +/- 0.0287	0.0834 +/- 0.0017
10.0	-0.4782 +/- 0.0464	-0.0829 +/- 0.0304	0.0740 +/- 0.0020

Table 4. Adaptive attacker at fixed D2, mean +/- std over seeds 42, 7, 123 (n = 3). Uniform trust is 0.10; lambda = 0 is the ordinary attacker from Table 2. Produced by adaptive_attacker.py.

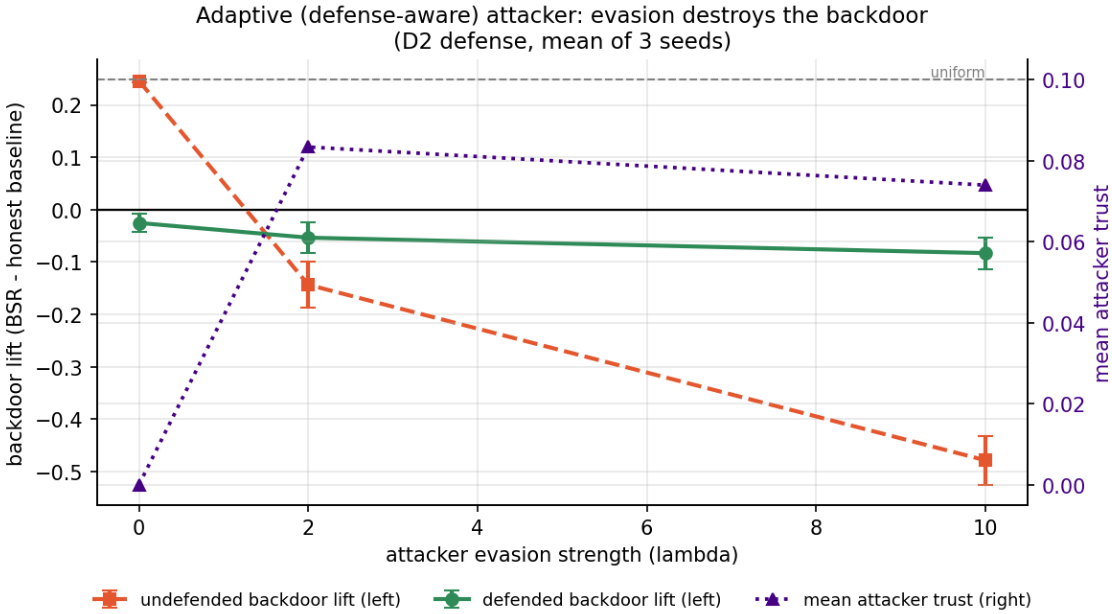

Figure 2. Defended backdoor lift (green, left axis) and mean attacker trust (purple, right axis) versus evasion strength. As the attacker evades better (trust rises toward uniform), the backdoor collapses; defended lift stays negative throughout.

Insight

- The adaptive attacker cannot win. Turning up evasion does raise its trust from 0.0001 toward uniform (0.083 at lambda = 2), so it genuinely hides better, but its undefended backdoor lift collapses from +0.2457 to -0.14 and then -0.48 at lambda = 10.
- The reason is structural, not luck: the CN0 backdoor needs the model to call CN0-benign-high inputs benign, while the CN0 probe rewards calling those same inputs spoofed. Evading the probe and keeping the backdoor are directly opposed, so training for one destroys the other.

There is no evasion setting where the attacker both hides from the trust score and keeps a working backdoor. The layered defended lift is negative at every lambda. This closes the adaptive-attacker question, the strongest open threat to a behavioral-trust defense.

5. Scope and future work

The core research questions are answered: the attack works and inflation worsens it (Table 2); each defense layer contributes and only the layered defense drives lift negative while preserving utility (Table 2); the defense generalizes across the discriminative feature set including a mixed trigger (Table 3); and a defense-aware adaptive attacker cannot both evade the probe and keep a working backdoor (Table 4). What remains is deliberately outside the scope of this study, not unfinished within it.

- Base detector accuracy. Honest spoofing recall is about 0.53 on this simplified public dataset. This is a property of the dataset and the small model, not the defense, and is why backdoor lift against each seed's own honest baseline is the primary metric. A stronger detector would not change any defense conclusion.
- Non-IID and larger fleets. Ten clients, two attackers, IID partitions. Non-IID data would widen the honest trust spread and is the setting most likely to stress the false-positive rate; it is the natural next study, and the IID choice is a stated scope boundary consistent with the Byzantine-robust FL literature.
- Seed count. Three seeds are enough to separate the attack effect from noise and to report mean and standard deviation, but not to resolve sub-noise differences between neighbouring defended variants, and we claim no such difference.
- Triggers outside the probe set. PQP and PIP carry near-zero class-discriminative signal (Cohen's d below 0.05) and are excluded from probing by design, so they are not tested as triggers. This is a scope definition, not a gap: the claim is trigger-agnostic within the discriminative feature set.

None of these affects the claims the paper makes. They mark where the contribution ends rather than work left undone.